9.6 Alternating Series, Absolute and Conditional Convergence

Alternating Series: terms alternate between positive and negative

Examples:

1.) $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \ldots + \frac{(-1)^{n+1}}{n} + \ldots$ alternating harmonic series, convergent

2.) $-2 + 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \ldots + \frac{(-1)^n 4}{2^n} + \ldots$ $r = -\frac{1}{2}$, converges to $-\frac{4}{3}$

3.) $1 - 2 + 3 - 4 + 5 - 6 + \ldots + (-1)^{n+1} n + \ldots$ divergent

The first series is called **alternating harmonic series**, and it is a convergent series.

The second series is a geometric series with ratio $-\frac{1}{2}$, and so converges to $\frac{-2}{1-(-\frac{1}{2})} = -\frac{4}{3}$.

The third series diverges, because the terms do not approach zero. $\Rightarrow$ fails the nth term test.

Theorem (The Alternating Series Test - Leibniz's Test):

The series $\sum_{n=1}^{\infty}(-1)^{n+1}u_n = u_1 - u_2 + u_3 - u_4 + \ldots\ldots$ converges if all three of the following conditions are satisfied:

1. The u_n's are all positive. ✓ (only apply if it is an alternating series).
2. The positive u_n's are (eventually) decreasing: $u_n \geq u_{n+1}$ for all $n \geq N$, for some integer N.
3. $u_n \to 0$. (otherwise, fails the nth term test)

Example: The alternating harmonic series $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n}$ satisfy all the three conditions, and hence converges.

- alternating ✓
- $\frac{1}{n} \to 0$ ✓
- $\frac{1}{n}$ decreasing.

Example: Consider the alternating series $\sum_{n=1}^{\infty}(-1)^{n+1}u_n$, where $u_n = \frac{10n}{n^2+16}$.

- $u_n > 0$
- $u_n \to 0$? $\quad u_n = \frac{10}{n + 16/n} \to 0$ as $n \to \infty$.
- Check condition 2: $f(x) = \frac{10x}{x^2+16}$

$$f'(x) = \frac{10(x^2+16) - 20x^2}{(x^2+16)^2}$$

$$= \frac{10(16 - x^2)}{(x^2+16)^2} \leq 0 \text{ for } x \geq 4$$

So the sequence $\{u_n\}$ is decreasing for $n \geq 4$.

So the series $\sum_{n=1}^{\infty}(-1)^{n+1}u_n$ converges by the AST.

But what about $\sum_{n=1}^{\infty}u_n$? Divergent series: Limit Comparison w/ $\sum \frac{1}{n}$.

Proof of Leibniz's Test: (Assume $N = 1$)

For n even, $n = 2m$, the partial sum

$$S_n = S_{2m} = \underbrace{(u_1 - u_2)}_{\geq 0} + \underbrace{(u_3 - u_4)}_{\geq 0} + \underbrace{(u_5 - u_6)}_{\geq 0} + \cdots + \underbrace{(u_{2m-1} - u_{2m})}_{\geq 0}$$

$\Rightarrow \{S_{2m}\}_{m \geq 1}$ is an increasing sequence.

($\{S_2, S_4, S_6, S_8, \ldots\}$)

$$S_{2m} = u_1 - \underbrace{(u_2 - u_3)}_{\geq 0} - \underbrace{(u_4 - u_5)}_{\geq 0} - \cdots - \underbrace{(u_{2m-2} - u_{2m-1})}_{\geq 0} - \underbrace{u_{2m}}_{\geq 0}$$

$\leq u_1$ for all m.

So $\{S_{2m}\}$ is an increasing sequence that is bounded above.

By Monotone Convergence theorem, $\lim_{m \to \infty} S_{2m} = L \ (< \infty)$.

For n odd, say $n = 2m+1$, $S_n = S_{2m} + u_{2m+1}$

$$\lim_{m \to \infty} S_{2m+1} = \lim_{m \to \infty} S_{2m} + \lim_{m \to \infty} u_{2m+1} = L + 0 = L.$$

$\Rightarrow \{S_n\}$ converges to L. $\square$

Example:

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\ln n}{n} = \ln 1 - \frac{\ln 2}{2} + \frac{\ln 3}{3} - \frac{\ln 4}{4} + \cdots$$

- $\frac{\ln n}{n} \to 0$: $\lim_{n\to\infty} \frac{\ln n}{n} = \lim_{x\to\infty} \frac{\ln x}{x} = \lim_{x\to\infty} \frac{1/x}{1} = 0$.

- $f(x) = \frac{\ln x}{x}$: $f'(x) = \frac{1 - \ln x}{x^2} < 0$ for $x > e$.

  so $\left\{\frac{\ln n}{n}\right\}$ is a decreasing sequence from $n = 3$ onwards.

By the AST, $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\ln n}{n}$ converges.

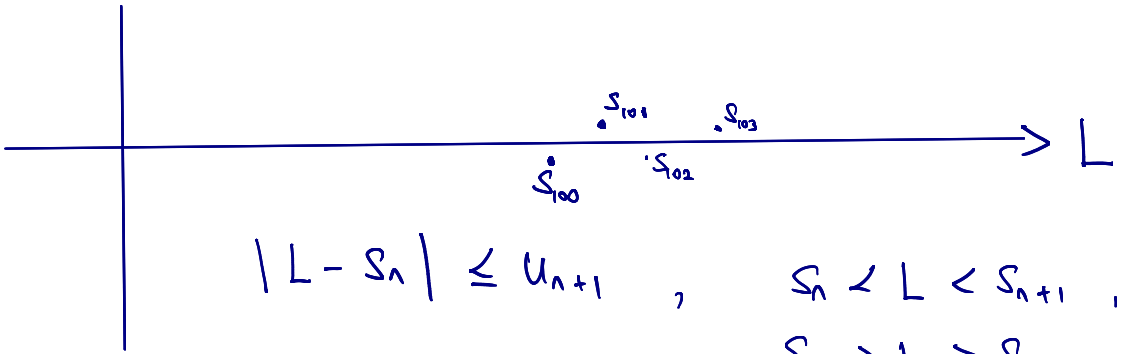
(Note that $\sum_{n=1}^{\infty} \frac{\ln n}{n}$ diverges.)

Theorem (The Alternating Series Estimation Theorem): If the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} u_n$ satisfies the three conditions of Leibniz's Test, then for $n \geq N$,

$$s_n = u_1 - u_2 + \ldots + (-1)^{n+1} u_n$$

approximates the sum L of the series with an error whose absolute value is less than u_{n+1}, the absolute value of the first unused term. Furthermore, the sum L lies between any two successive partial sums s_n and s_{n+1}, and the remainder $L - s_n$, has the same sign as the first unused term.

Suppose $\sum_{n=1}^{\infty} (-1)^{n+1} u_n = L \ (< \infty)$



$|L - S_n| \leq u_{n+1}$,

$S_n < L < S_{n+1}$, n even

$S_n > L > S_{n+1}$, n odd.

To estimate $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} = L$, we can compute S_{10}.

$$S_{10} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots - \frac{1}{10} \approx 0.6456$$

Then $$|L - S_{10}| < \frac{1}{11} \Rightarrow S_{10} - \frac{1}{11} < L < S_{10} + \frac{1}{11}$$

If require precision within 0.01, we can estimate L with S_{99}: $$|L - S_{99}| < \frac{1}{100}.$$

Absolute and Conditional Convergence

We have investigated the convergence of the following two series.

Harmonic Series: $\sum_{n=1}^{\infty} \frac{1}{n}$

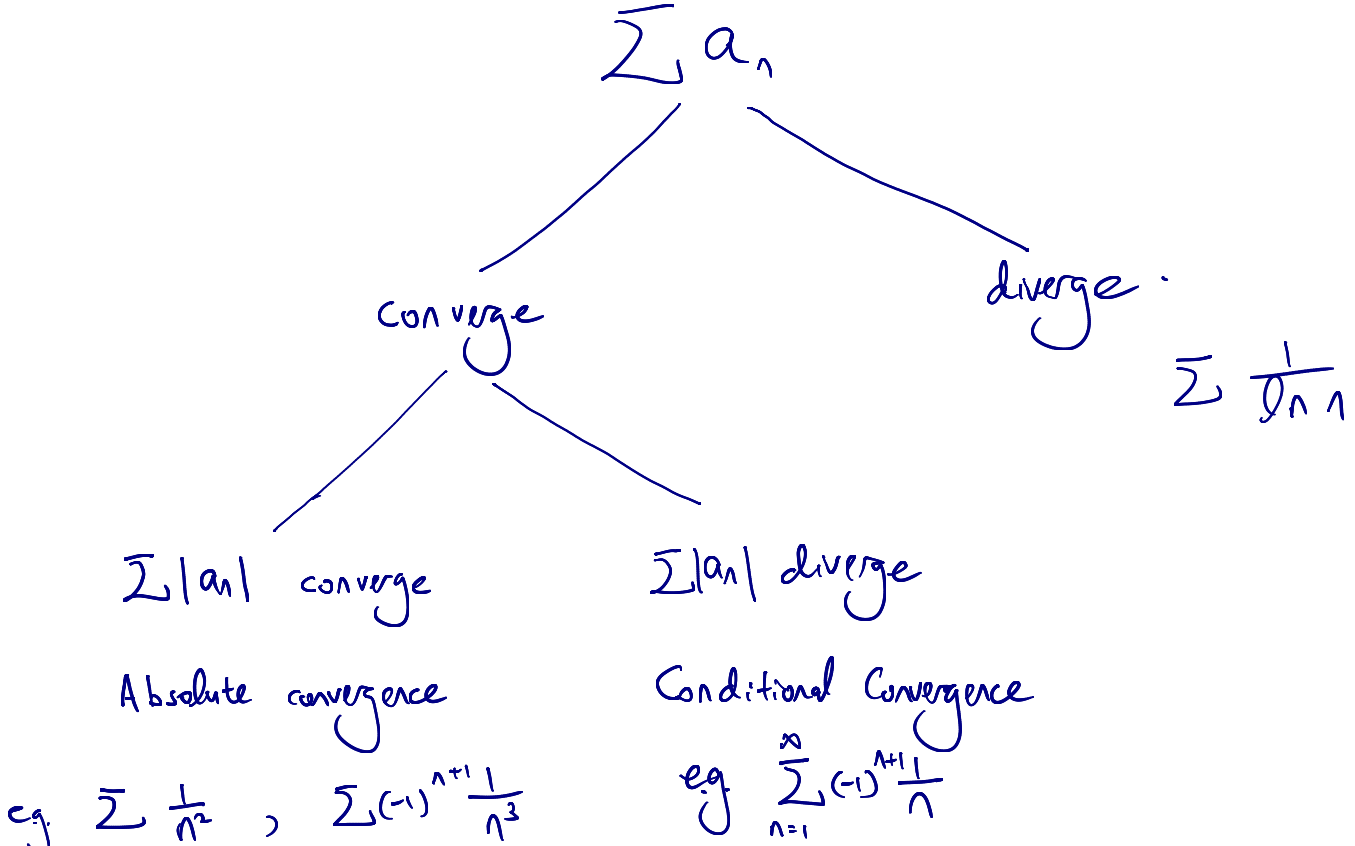
Alternating Harmonic Series: $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$

Observe that the absolute values of the terms in these two series are the same:

$\left|\frac{(-1)^{n+1}}{n}\right| = \frac{1}{n}$

These two series illustrate the fact that a series $\sum a_n$ may be convergent, but the series

$\sum |a_n|$ obtained by taking absolute values of the terms may be divergent. This observation leads to an important definition.



Absolute Convergence $\Rightarrow$ Convergence.

Definition: Let $\{a_n\}_{n=1}^{\infty}$ be a sequence. The series $\sum a_n$ **converges absolutely** (is **absolutely convergent**) if the series $\sum |a_n|$ converges. A series is said to be **conditionally convergent** if it is convergent, but it is not absolutely convergent.

$\sum a_n$ converges $\sum |a_n|$ diverges

Examples:

1. It is trivial that a series with positive terms is absolutely convergent if and only if it is a convergent series. If $a_n \geq 0$ then $|a_n| = a_n$ so $\sum |a_n| = \sum a_n$.
2. The alternating harmonic series is conditionally convergent.

$\sum (-1)^{n+1} \frac{1}{n}$ converges (by AST).

but $\sum \left|(-1)^{n+1} \frac{1}{n}\right| = \sum \frac{1}{n}$ diverges.

Theorem (The Absolute Convergence Test): If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

Caution: The converse is not true.

Proof: Observe that $-|a_n| \le a_n \le |a_n|$

$\Rightarrow \quad 0 \le |a_n| + a_n \le 2|a_n|$.

If $\sum |a_n|$ converges, then $\sum_{n=1}^{\infty} 2|a_n|$ converges too.

So by DCT, $\sum (|a_n| + a_n)$ converges

Note that $\sum a_n = \left(\sum [a_n + |a_n|]\right) - \left(\sum |a_n|\right)$

converges converges

$\Rightarrow \sum a_n$ converges.

Examples:

1. $\sum_{n=1}^{\infty} \underbrace{(-1)^{n+1}\frac{1}{n^2}}_{a_n}$: $\sum |a_n| = \sum \frac{1}{n^2}$ converges ($\because$ 2-series)

$\Rightarrow \sum_{n=1}^{\infty} (-1)^{n+1}\frac{1}{n^2}$ converges too.

2. $\sum_{n=1}^{\infty} \underbrace{\frac{\sin n}{n^2}}_{a_n}$ $\sum |a_n| = \sum \left|\frac{\sin n}{n^2}\right| = \sum \frac{|\sin n|}{n^2}$

By comparing with $\sum \frac{1}{n^2}$ $\left(0 \le \frac{|\sin n|}{n^2} \le \frac{1}{n^2}\right)$,

we deduce that $\sum \left|\frac{\sin n}{n^2}\right|$ converges.

Thus by the Absolute Convergence Test, $\sum \frac{\sin n}{n^2}$ converges too.

3. Let p be a positive constant. Consider the alternating p-series $\sum_{n=1}^{\infty} \boxed{\dfrac{(-1)^{n+1}}{n^p}}$. ($a_n$)

$\sum |a_n| = \sum \frac{1}{n^p}$

For $p > 1$: $\sum |a_n|$ converges

$\Rightarrow \sum a_n = \sum (-1)^{n+1}/n^p$ converges.

For $0 < p \leq 1$: $\sum |a_n| = \sum \frac{1}{n^p}$ diverges.

$\sum_{n=1}^{\infty} (-1)^{n+1}/n^p$ converges by Alternating series test.

So when $0 < p \leq 1$, $\sum_{n=1}^{\infty} (-1)^{n+1}/n^p$ is <u>conditionally convergent</u>.

(when $p \leq 0$, the series clearly diverges) -

Rearranging Series

We can always rearrange the terms of an infinite series that is absolutely convergent. But we cannot do so for conditionally convergent series.

Theorem (The Rearrangement Theorem for Absolutely Convergent Series): If $\sum_{n=1}^{\infty} a_n$ converges absolutely, and $b_1, b_2, \ldots, b_n, \ldots$ is any arrangement of the sequence $\{a_n\}$, then

$\sum b_n$ converges absolutely and

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} a_n.$$

When $\sum a_n$ converges conditionally, say $\sum a_n = L$.

If $b_1, b_2, \ldots$ are rearrangement of $\{a_n\}$, then $\sum b_n \neq \sum a_n$.

A remarkable observation due to Riemann: If $\sum a_n$ is a conditionally convergent series, and if r is an arbitrary number, then there is a rearrangement of $\sum a_n$ that converges to r.

Example: We know the alternating harmonic series $\sum \frac{(-1)^{n+1}}{n}$ converges to some number L. $(L \neq 0)$

$$2L = 2\sum_{n=1}^{\infty} (-1)^{n+1}/n = 2\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \frac{1}{9} - \frac{1}{10} + \cdots\right)$$

$$= 2 - 1 + \frac{2}{3} - \frac{1}{2} + \frac{2}{5} - \frac{1}{3} + \frac{2}{7} - \frac{1}{4} + \frac{2}{9} - \frac{1}{5} + \cdots$$

$$= (2-1) - \frac{1}{2} + \left(\frac{2}{3} - \frac{1}{3}\right) - \frac{1}{4} + \left(\frac{2}{5} - \frac{1}{5}\right) - \frac{1}{6} + \cdots$$

$$= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots$$

$$= L$$

$\Rightarrow 2L = L$ ✗ as $L \neq 0$

$\therefore$ We cannot rearrange the series as we like.

$$\sum_{n=1}^{\infty} \boxed{(-1)^{n+1} \frac{2n^2(\ln n) + n(\ln n)^2}{7n^3(\ln n)^3 + 4}} \quad a_n$$

converges ← Absolute CT.

If this converges, then

$$\sum_{n=1}^{\infty} |a_n| = \sum \boxed{\frac{2n^2(\ln n) + n(\ln n)^2}{7n^3(\ln n)^3 + 4}} \quad = f(n)$$

$$\approx \frac{n^2(\ln n)}{n^3(\ln n)^3} = \frac{1}{n(\ln n)^2}$$

Let $g(n) = \frac{1}{n(\ln n)^2}$, we check that $\lim_{n \to \infty} \frac{f(n)}{g(n)} = \frac{2}{7}$

By the LCT, $\sum f(n)$ converges $\Leftrightarrow$ $\sum g(n)$ converges.

To check the convergence of $\sum \frac{1}{n(\ln n)^2}$, we use integral test.

So we check if $\int_{\ln 2}^{\infty} \frac{1}{x(\ln x)^2}\, dx$ converges. This improper integral converges to $\frac{1}{\ln 2}$.

Summary of Tests to Determine Convergence or Divergence

- No fast rules on which test to apply to a given series.
- There are other tests we have not covered in the course.

1. **The *n*th-Term Test for Divergence:** Unless $a_n \to 0$, the series diverges.
2. **Geometric series:** $\sum ar^n$ converges if $|r| < 1$; otherwise it diverges.
3. ***p*-series:** $\sum 1/n^p$ converges if $p > 1$; otherwise it diverges.
4. **Series with nonnegative terms:** Try the Integral Test or try comparing to a known series with the Direct Comparison Test or the Limit Comparison Test. Try the Ratio or Root Test.
5. **Series with some negative terms:** Does $\sum |a_n|$ converge by the Ratio or Root Test, or by another of the tests listed above? Remember, absolute convergence implies convergence.
6. **Alternating series:** $\sum a_n$ converges if the series satisfies the conditions of the Alternating Series Test.